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C23-M-303

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BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DME – THIRD SEMESTER EXAMINATION

THERMAL ENGINEERING—I

Time : 3 hours] [*Total Marks* : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Assume for Air, R = 0.287 kJ/kg/K and γ=1.4

1. Define System and List out different type of systems.

2. Differentiate between reversible and irreversible processes.

3. State Avogadro's law and Regnault's law.

4. Calculate the value of gas constant of gas whose molecular weight is 28.

5. Draw P-V and T-S diagrams for Isentropic Process.

6. Why Isothermal process is called as hyperbolic process?

7. What is the difference between actual engine and engine work on air standard cycle?

8. Represent Otto cycle on P-V and T-S diagrams.

9. Define combustion, reactant and product of combustion.

10. Write Dulong's formula and describe each term.

PART—B

10×5=50

Instructions : (1) Answer *any* **five** questions.

(2) Each question carries **ten** marks.

(3) Assume for Air, R = 0.287 kJ/kg/K and $\gamma = 1.4$

11. A system undergoes a cycle consisting of the three processes listed in the table.

Compute the missing values (a, b, c, d). All quantities are in kJ.

Process	Heat transfer	Work transfer	Change in Internal Energy
1 – 2	a	100	100
2 – 3	b	–50	c
3 – 1	100	d	–200

12. *(a)* State and explain Kelvin-Planck statement of second law. 5

(b) Derive characteristic gas equation. 5

13. An air of 3 kg expanded from 12 bar and 2 m^3 to 2 bar and 10 m^3. During this expansion, heat supplied is 35 kJ and work output is 90 kJ. Calculate change in internal energy and state whether it is gain or loss. Also find temperatures at beginning and end of the process. Assume suitable data for air.

14. A cylinder contains 0.45 m^3 of a gas at 1×10^5 N/m^2 and 80 °C. The gas is compressed to a volume of 0.13 m^3, the final pressure being 5×10^5 N/m^2. Assume suitable data for air. Determine —

(i) the mass of gas;

(ii) the value of index 'n' for compression;

(iii) the increase in internal energy of the gas;

(iv) the heat received or rejected by the gas during compression.

15. Certain mass of air at 180 °C expands adiabatically to three times its original volume and during the process, there is a fall in temperature to 15 °C. Assume suitable data for air. The work done during the process is 52.5 kJ. Calculate mass, change in internal energy, volumes at beginning and end of expansion.

16. A diesel cycle has a compression ratio of 18 and the heat transferred to the working fluid per cycle is 1800 kJ/kg. At the beginning of the compression stroke, the pressure is 1 bar and the temperature is 300 K. Assume suitable data for air. Calculate air standard efficiency.

17. *(a)* Compare Otto cycle with diesel cycle. 5

(b) State the advantages and disadvantages of commonly used liquid fuels. 5

18. The percentage composition of sample of liquid fuel by weight is C = 84.8 percent and H_2 = 15.2 percent. Calculate *(i)* the weight of air needed for the combustion of 1 kg of fuel; *(ii)* the volumetric composition of the products of combustion if 15 percent excess air is supplied.

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